Unit 3

Lesson **Four**

Applications on definite Integral

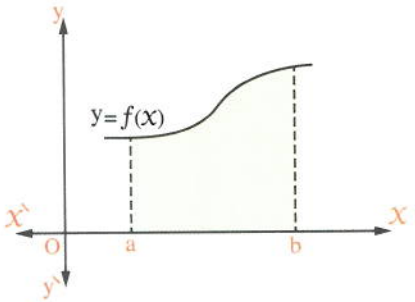
First Areas in the plane

The area of a region bounded by the curve of the function f and X-axis in the interval [a , b]

⊙ If a function f is continuous on the interval [a , b] and A is the area of the region bounded by the curve of the function f and X-axis and the two straight lines $X = a$, $X = b$ and if :

1. $f(X) \geq 0$

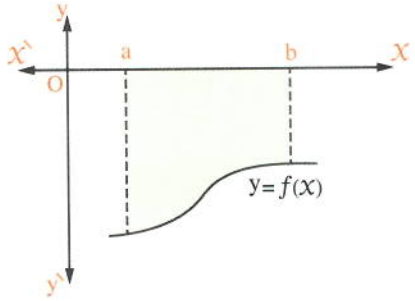
i.e. "The region above X-axis"



, then $A = \int_a^b f(X)\, dX$

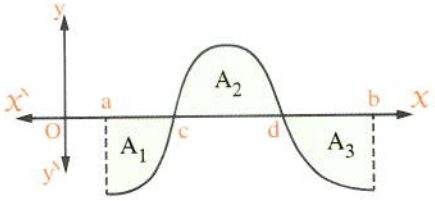
2. $f(X) \leq 0$

i.e. "The region below X-axis"



, then $A = -\int_a^b f(X)\, dX = \left|\int_a^b f(X)\, dX\right|$

⊙ If the curve of the function f intersects the X-axis at $X = c$ and $X = d$ where c and d belong to [a , b] as in the opposite figure :



We find that : $f(X) \geq 0$ for all $X \in [c , d]$

, $f(X) \leq 0$ for all $X \in [a , c]$ and $X \in [d , b]$

$\therefore$ The shaded area $(A) = A_1 + A_2 + A_3$

i.e. $A = \left|\int_a^c f(X)\, dX\right| + \int_c^d f(X)\, dX + \left|\int_d^b f(X)\, dX\right|$

Notice that

The absolute value to the two areas A_1 , A_3 because they are below X-axis.

Remarks

1. It is favorable to graph the curve of the function to identify the region above or below X-axis.
2. Sometimes it is difficult to sketch some curves then , its favorable to find zeroes of the function (even the limits of the integral are given) which divides the domain of the function $[a , b]$ if exist into intervals and determine the sign of the function in each part and so you know if the region is above or below X-axis.
3. The value of the integration , could be positive or negative but the area is always positive.
4. In general , the area of the included region between any continuous function : $y = f(X)$ and th. X-axis and the two straight lines $X = a$, $X = b$

   is $A = \int_a^b |f(X)| \, dX$

Example 1

Find the shaded area in each of the following :

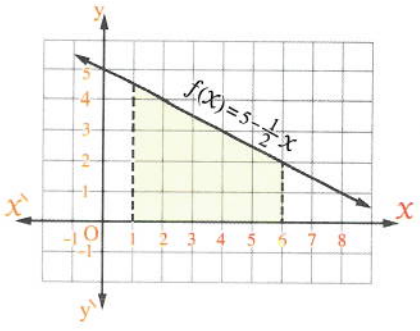


Figure (1)

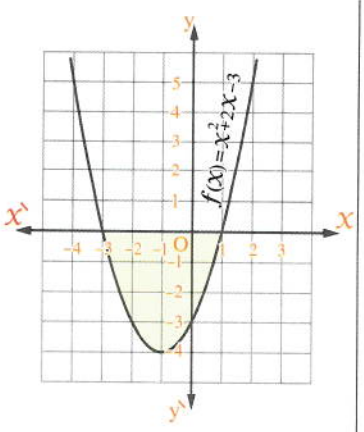


Figure (2)

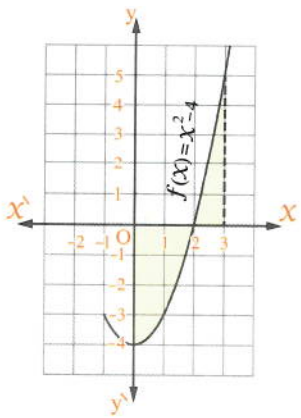


Figure (3)

Solution

Figure (1) :

$\because$ The shaded region lies above X-axis

$\therefore$ The area of the shaded region $= \int_1^6 f(X)\,dX = \int_1^6 \left(5 - \frac{1}{2}X\right) dX$

$= \left[5X - \frac{1}{4}X^2\right]_1^6 = (30 - 9) - (5 - \frac{1}{4}) = 16\ \frac{1}{4}$ square unit.

Figure (2) :

$\because$ The shaded region lies below X-axis

$\therefore$ The area of the shaded region $= \left|\int_{-3}^1 f(X)\,dX\right|$

$= \left|\int_{-3}^1 (X^2 + 2X - 3)\,dX\right| = \left|\left[\frac{1}{3}X^3 + X^2 - 3X\right]_{-3}^1\right|$

$= \left|\left(\frac{1}{3} + 1 - 3\right) - (-9 + 9 + 9)\right| = \left|\frac{-32}{3}\right| = \frac{32}{3}$ square unit.

Figure (3) :

$\because$ The shaded region lies below X-axis for every $X \in [0\ ,\ 2]$

and lies above X-axis for every $X \in [2\ ,\ 3]$

$\therefore$ The area of the shaded region $= \left|\int_0^2 f(X)\,dX\right| + \int_2^3 f(X)\,dX$

$= \left|\int_0^2 (X^2 - 4)\,dX\right| + \int_2^3 (X^2 - 4)\,dX$

$= \left|\left[\frac{1}{3}X^3 - 4X\right]_0^2\right| + \left[\frac{1}{3}X^3 - 4X\right]_2^3$

$= \left|\frac{8}{3} - 8\right| + \left(\left(9 - 12\right) - \left(\frac{8}{3} - 8\right)\right) = \frac{23}{3}$ square unit.

Example 2

Find the area of the region bounded by the curve of the function $f : f(X) = e^X$ and the two straight lines $X = 0$, $X = \ln 3$ and X-axis

Solution

$\because$ The function is continuous on the interval $[0\ ,\ \ln 3]$

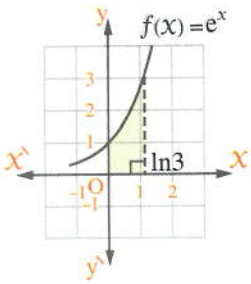
, $\because f(X) = e^X > 0$, for every $X \in [0\ ,\ \ln 3]$

$\therefore A = \int_0^{\ln 3} e^X\,dX = \left[e^X\right]_0^{\ln 3}$

$= e^{\ln 3} - e^0 = 3 - 1 = 2$ square unit.

Remember

$e^{\ln a} = a$



Example 3

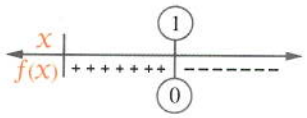
Find the area of the region bounded by the curve of the function $y = 1 - x^3$ and the straight lines $y = 0$, $x = -1$, $x = 2$

Solution

* Put $y = 0$ $\quad\therefore 1 - x^3 = 0$

$\therefore x^3 = 1$ $\quad\therefore x = 1$

* Investigate the sign of the function $f : f(x) = 1 - x^3$



$\therefore f(x) \geq 0$ for every $x \in [-1 , 1]$

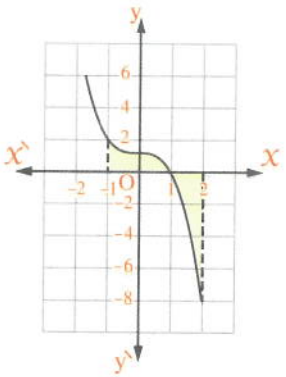
, $f(x) \leq 0$ for every $x \in [1 , 2]$

$\therefore A = \int_{-1}^{1} y \, dx + \left| \int_{1}^{2} y \, dx \right|$

$= \int_{-1}^{1} (1 - x^3) \, dx + \left| \int_{1}^{2} (1 - x^3) \, dx \right|$

$= \left[x - \frac{1}{4} x^4 \right]_{-1}^{1} + \left| \left[x - \frac{1}{4} x^4 \right]_{1}^{2} \right| = \left(\frac{3}{4} + \frac{5}{4} \right) + \left| -2 - \frac{3}{4} \right|$

$= 2 + 2\frac{3}{4} = 4\frac{3}{4}$ square unit.



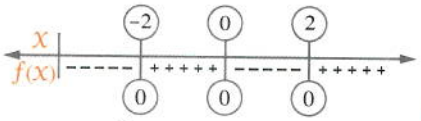
Example 4

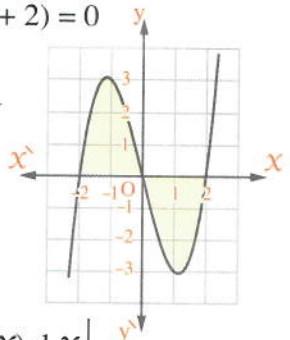
Find the area of the region bounded by the curve of the function $y = x^3 - 4x$ and the x-axis

Solution

* Put $y = 0$ $\quad\therefore x(x^2 - 4) = 0$ $\quad\therefore x(x - 2)(x + 2) = 0$

$\therefore x = 0$ or $x = 2$ or $x = -2$





* Study the sign of the function $f : f(x) = x^3 - 4x$

$\therefore f(x) \geq 0$ for every $x \in [-2 , 0]$, $f(x) \leq 0$ for every $x \in [0 , 2]$

$\therefore A = \int_{-2}^{0} y \, dx + \left| \int_{0}^{2} y \, dx \right| = \int_{-2}^{0} (x^3 - 4x) \, dx + \left| \int_{0}^{2} (x^3 - 4x) \, dx \right|$

$= \left[\frac{1}{4} x^4 - 2x^2 \right]_{-2}^{0} + \left| \left[\frac{1}{4} x^4 - 2x^2 \right]_{0}^{2} \right|$

$= 4 + 4 = 8$ square unit.

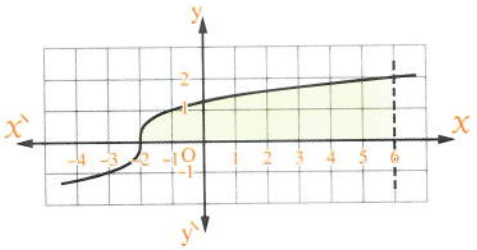
Example 5

Find the area of the region bounded by the curve of the function $f : f(x) = \sqrt[3]{x+2}$ and the straight line $x = 6$, and above x-axis.

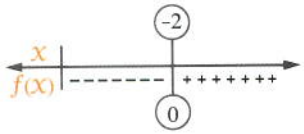
Solution

* Put $f(x) = 0$ $\quad\therefore \sqrt[3]{x+2} = 0$

$\therefore x + 2 = 0 \quad\therefore x = -2$



* Investigate the sign of the function



$\therefore f(x) \geq 0$ for every $x \in [-2 , 6]$

$\therefore$ The area $= \int_{-2}^{6} f(x)\, dx = \int_{-2}^{6} (x+2)^{\frac{1}{3}}\, dx$

$= \left[\frac{3}{4}(x+2)^{\frac{4}{3}}\right]_{-2}^{6} = \frac{3}{4}(8)^{\frac{4}{3}} - 0 = 12$ square unit.

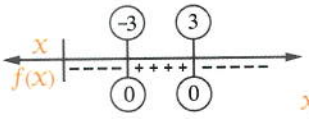
Example 6

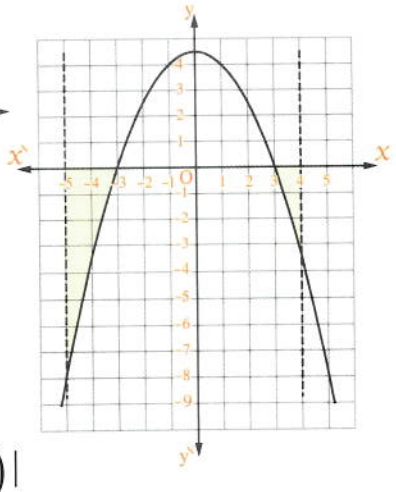
Find the area of the region bounded by the curve of the function $y = \frac{1}{2}(9 - x^2)$ and the straight lines $y = 0$, $x = -5$, $x = 4$ and lies below the x-axis.

Solution

* Put $y = 0 \quad\therefore 9 - x^2 = 0 \quad\therefore x = \pm 3$

* Investigate the sign of the function





$\therefore$ The area $= \left|\int_{-5}^{-3} y\, dx\right| + \left|\int_{3}^{4} y\, dx\right|$

$= \left|\int_{-5}^{-3} \frac{1}{2}(9 - x^2)\, dx\right| + \left|\int_{3}^{4} \frac{1}{2}(9 - x^2)\, dx\right|$

$= \left|\left[\frac{9}{2}x - \frac{1}{6}x^3\right]_{-5}^{-3}\right| + \left|\left[\frac{9}{2}x - \frac{1}{6}x^3\right]_{3}^{4}\right|$

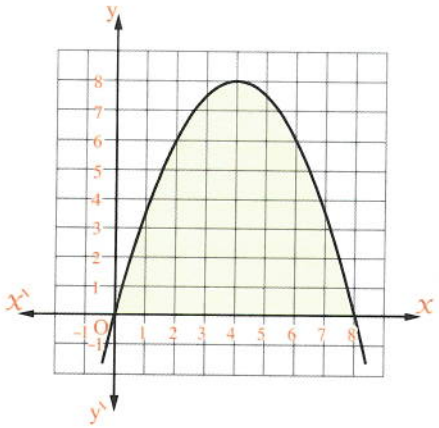
$= \left|\left(\frac{9}{2}(-3) - \frac{1}{6}(-3)^3\right) - \left(\frac{9}{2}(-5) - \frac{1}{6}(-5)^3\right)\right|$

$+ \left|\left(\frac{9}{2}(4) - \frac{1}{6}(4)^3\right) - \left(\frac{9}{2}(3) - \frac{1}{6}(3)^3\right)\right| = 9$ square unit.

Example 7

An engineer design a hotel reception in form of the curve whose equation $y = 4x - \frac{1}{2}x^2$ where x is in metres. If it is covered by glass , the cost of one square metre is L.E. 1200 Find the cost of the glass.

Solution



* Put $y = 0$ $\quad\therefore 4x - \frac{1}{2}x^2 = 0$

$\therefore \frac{1}{2}x(8 - x) = 0$ $\quad\therefore x = 0$ or $x = 8$

$\therefore$ The area (A) $= \int_0^8 (4x - \frac{1}{2}x^2)\, dx$

$= \left[2x^2 - \frac{1}{6}x^3\right]_0^8$

$= \left[2(8)^2 - \frac{1}{6}(8)^3\right] - [\text{zero}]$

$= \frac{128}{3}$ square metre.

$\therefore$ The cost = the area × the cost of one square metre

$= \frac{128}{3} \times 1200 =$ L.E. 51200

Example 8

By using the area under the curve of the function $y = \sqrt{16 - x^2}$, find $\int_0^4 \sqrt{16 - x^2}\, dx$

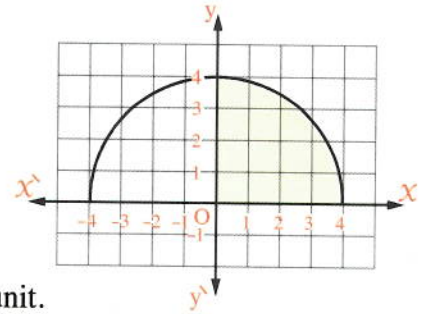
Solution

$\because$ The function $y = \sqrt{16 - x^2}$ is continuous and not negative on $[0 , 4]$, $y = \sqrt{16 - x^2}$ represents the part above x-axis from the circle resulted from squaring both sides.

i.e. $y^2 = 16 - x^2$ $\quad\therefore x^2 + y^2 = 16$

and it is a circle whose centre $(0 , 0)$ and its raduis length = 4 units as shown in the figure.

$\therefore \int_0^4 \sqrt{16 - x^2}\, dx =$ the shaded area bounded by the curve of the function and the x-axis and the straight line $x = 0$, $x = 4$



$= \frac{1}{4} \times$ the area of the circle $= \frac{1}{4} \times \pi r^2$

$= \frac{1}{4} \times \pi \times 4^2 = 4\pi$ square unit.

Second Volumes of revolution solids

Revolution Solid

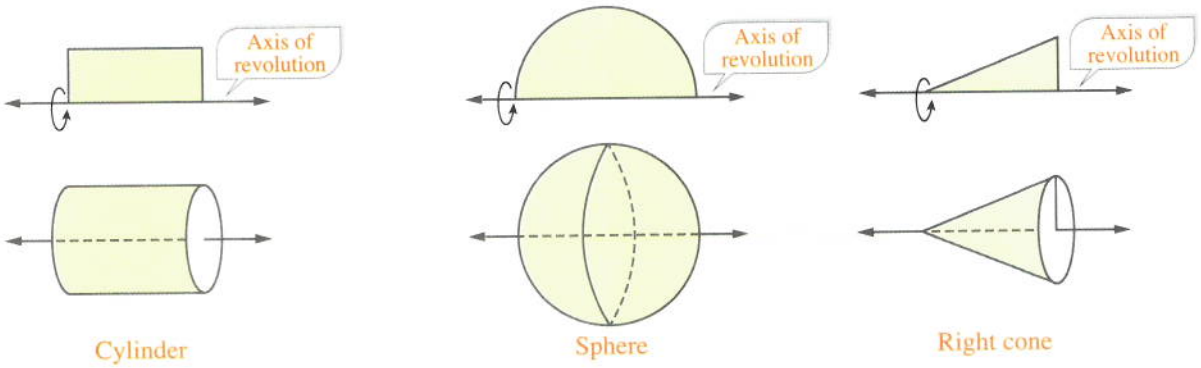
* It is a solid generated by revolving a plane area a complete revolution about a fixed straight line in its plane called «axis of revolution»

For example :

1. If a rectangular region revolves about one of its sides a complete revolution , the produced solid is cylinder.

2. If a semi-circluar region revolves about its diameter a complete revolution , the produced solid is a sphere.

3. If a right-angled triangular region revolves about one side of its right angle , the produced solid is a circular right cone and the following figures show that :

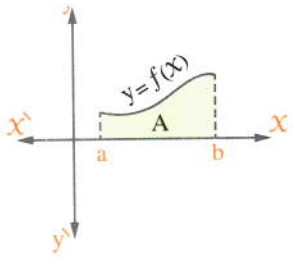


Cylinder Sphere Right cone

The volume of a solid produced by revolving a region about X-axis :

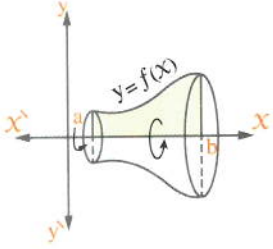
Assume f is a continuous function on the interval $[a , b]$, $f(X) \geq 0$ for every $X \in [a , b]$

* If the plane region (A) bounded by the curve $y = f(X)$ and X-axis and the two straight lines $X = a$, $X = b$ revolves a complete revolution about «X-axis»



Assuming that v is the volume of the solid of revolution

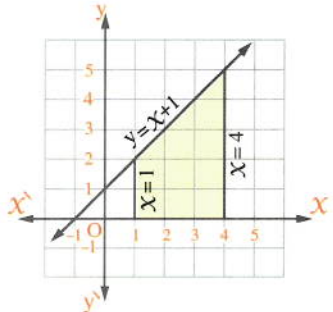
$$v = \pi \int_a^b y^2 \, dX = \pi \int [f(X)]^2 \, dX$$



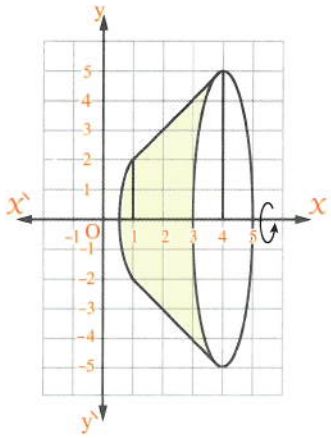
Example 9

Find the volume of the solid generated by revolving the region bounded by the straight lines : $y = x + 1$, $y = 0$, $x = 1$, $x = 4$ a complete revolution about x-axis.

Solution



«The region bounded by the straight lines»



«The shape of the produced solid of revolution»

$\because$ The function is continuous on the interval $[1 , 4]$, $f(x) \geq 0$ for all $x \in [1 , 4]$

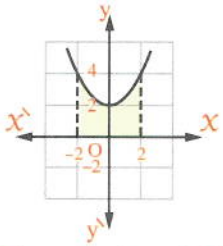
$\therefore$ The volume of the solid of revolution $(v) = \pi \int_1^4 y^2 \, dx = \pi \int_1^4 (x + 1)^2 \, dx$

$$= \pi \left[\frac{(x+1)^3}{3}\right]_1^4 = 39\,\pi \text{ cubic unit.}$$

Example 10

Find the volume of the solid generated by revolving the region bounded by the curve : $y = x^2 + 2$ and the x-axis and the two straight lines $x = -2$, $x = 2$ a complete revolution about x-axis.

Solution



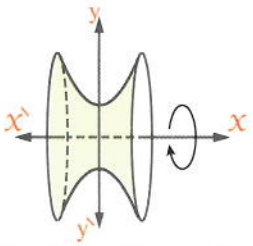
«The region bounded by the curve and the straight lines»

$\because$ The function is a continuous polynomial on the interval $[-2 , 2]$, $f(x) \geq 0$ for all $x \in [-2 , 2]$

$\therefore v = \pi \int_{-2}^2 y^2 \, dx = \pi \int_{-2}^2 (x^2 + 2)^2 \, dx$

$$= \pi \int_{-2}^2 (x^4 + 4x^2 + 4) \, dx$$

$$= \pi \left[\frac{1}{5}x^5 + \frac{4}{3}x^3 + 4x\right]_{-2}^2 = \frac{752}{15}\,\pi \text{ cubic unit.}$$



«The shape of the produced solid of revolution»

Example 11

Find the volume of the solid generated by revolving the region bounded by the curve : $y = 4 - x^2$ and the x-axis a complete revolution about x-axis.

Solution

To find the intersection points of the curve with the x-axis , put $4 - x^2 = 0$

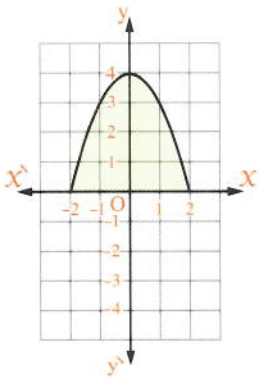
$\therefore x^2 = 4$ $\quad\quad$ $\therefore x = \pm 2$

$$\therefore v = \pi \int_{-2}^{2} y^2 \, dx = \pi \int_{-2}^{2} (4 - x^2)^2 \, dx$$

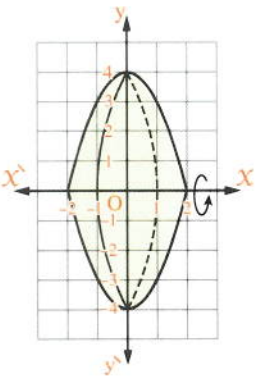
$$= \pi \int_{-2}^{2} (16 - 8x^2 + x^4) \, dx$$

$$= \pi \left[16x - \frac{8x^3}{3} + \frac{x^5}{5} \right]_{-2}^{2}$$

$$= \frac{512}{15} \pi \text{ cubic unit.}$$



«The region bounded by the straight lines»



«The shape of the produced solid of revolution»

Example 12

Find the volume of the solid generated by revolving the region bounded by the curve : $y = \sqrt{x + 4}$ and the two axis of coordinates a complete revolution About x-axis.

Solution

* To find the intersection points of the curve with the x-axis , put $y = 0$

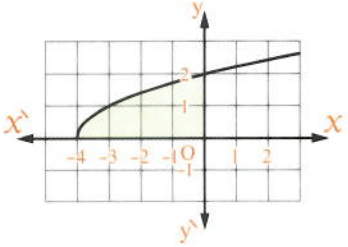
$\therefore x + 4 = 0$ $\quad\quad$ $\therefore x = -4$

* To find the intersection points of the curve with the y-axis , put $x = 0$

$y = \sqrt{4} = 2$

$$\therefore v = \pi \int_{-4}^{0} y^2 \, dx = \pi \int_{-4}^{0} (x + 4) \, dx$$

$$= \pi \left[\frac{1}{2}x^2 + 4x \right]_{-4}^{0} = 8\pi \text{ cubic unit.}$$



Remarks

By drawing the two curves $y = \sqrt{x}$ [figure (1)] , $y^2 = x$ [figure (2)] and by shading the region bounded by the curve and x-axis and $x = 2$ for example [as in figure (1)] and also by shading the region bounded by the curve and the straight line $x = 2$ for example [as in figure (2)] we notice that :

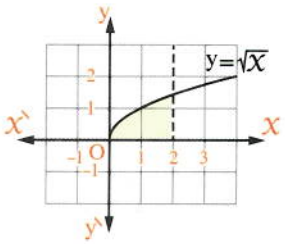


Fig. (1)

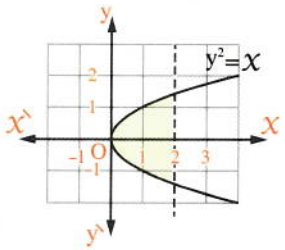


Fig. (2)

1. The area of the region [in figure (1)] = $\frac{1}{2}$ the area of the region [in figure (2)]
2. The volume of the solid of revolution produced by rotation the shaded region in [figure (1)] a complete revolution about x-axis equals the volume of the solid of revolution produced by rotation the shaded region in [figure (2)] half revolution about x-axis.

Example 13

Find the volume of the solid generated by revolving the region bounded by the curve $\frac{x^2}{c^2} + \frac{y^2}{d^2} = 1$ where c and d are two constants a complete revolution About the x-axis.

Solution

* To find the intersection points of the curve with the x-axis , put $y = 0$

$\therefore x^2 = c^2$ $\qquad\qquad$ $\therefore x = \pm c$

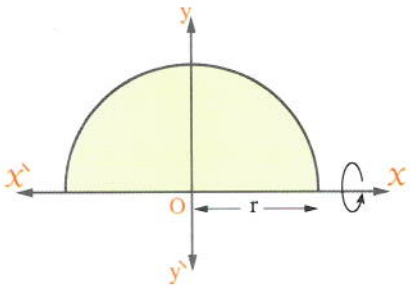
$$\therefore v = \pi \int_{-c}^{c} y^2 \, dx = \pi \int_{-c}^{c} d^2 \left(1 - \frac{x^2}{c^2}\right) dx = \pi d^2 \left[x - \frac{x^3}{3c^2}\right]_{-c}^{c}$$

$$= \pi d^2 \left[\left(c - \frac{c}{3}\right) - \left(-c + \frac{c}{3}\right)\right] = \frac{4}{3} \pi d^2 c$$

Example 14

Use the integration to find the volume of the sphere of radius length = r

Solution



Let a half circle of centre (0 , 0) and a radius length = r and its diameter coincide with the X-axis , then the equation of the half circle is

$X^2 + y^2 = r^2$ **i.e.** $y = \sqrt{r^2 - X^2}$

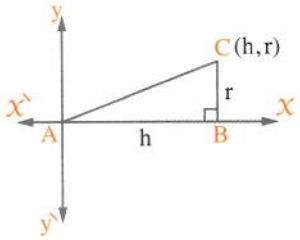
The sphere is produced by the revolving the plane region bounded by the half circle and the X-axis a complete revolution about X-axis.

$$\therefore v = \pi \int_{-r}^{r} y^2 \, dX = \pi \int_{-r}^{r} (r^2 - X^2) \, dX = \pi \left[r^2 X - \frac{1}{3} X^3 \right]_{-r}^{r} = \frac{4}{3} \pi r^3$$

Example 15

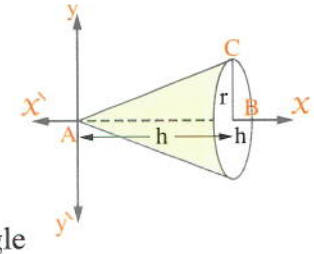
Use the integration to prove that volume of the right circular cone = $\frac{1}{3} \pi r^2 h$ where r is the radius length of its base and h is its height.

Solution



* We draw a right-angled triangle ABC where A is the origin point.
One of the sides of the right angle $\overline{AB}$ is coincided with the X-axis and the length of $\overline{AB} = h$, the length of $\overline{BC} = r$
, $\because$ A (0 , 0) , C (h , r)
$\therefore$ The equation of $\overleftrightarrow{AC}$ is $\frac{y - 0}{X - 0} = \frac{r - 0}{h - 0}$

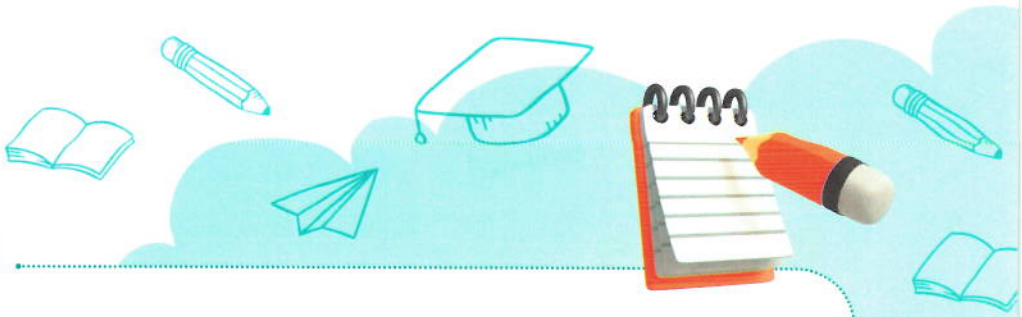
i.e. $\boxed{y = \frac{r}{h} X}$



* The right circular cone is produced by the revolving of the triangle ABC bounded by the straight line $\overleftrightarrow{AC}$ and the X-axis and the two straight lines $X = 0$, $X = h$

$$\therefore \text{The volume of the cone} = \pi \int_0^h y^2 \, dX = \pi \int_0^h \frac{r^2}{h^2} X^2 \, dX = \frac{\pi r^2}{h^2} \int_0^h X^2 \, dX$$

$$= \frac{\pi r^2}{h^2} \left[\frac{1}{3} X^3 \right]_0^h = \frac{\pi r^2}{h^2} \times \frac{1}{3} h^3 = \frac{1}{3} \pi r^2 h$$

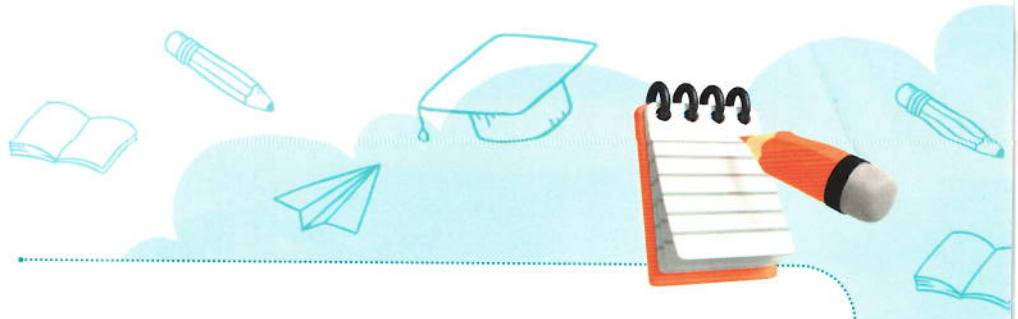


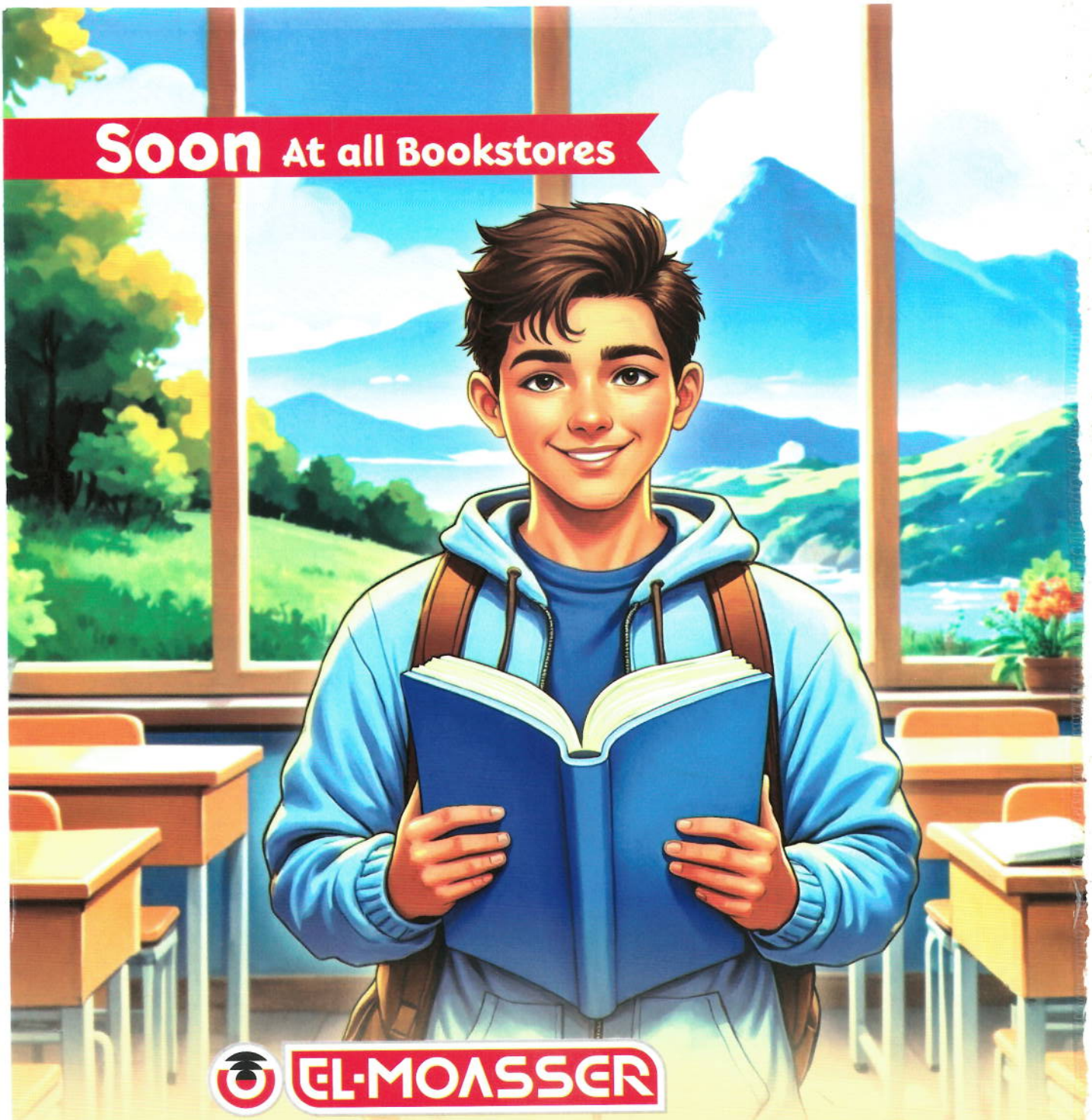
Notes



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3rd Sec.

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